

Developing socio-hydrology: Research progress, opportunities and challenges

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Abstract: The development of industrialization and urbanization has intensified the coupling of human activities and hydrological processes and promoted the emergence of socio-hydrology. This paper addresses the issue of socio-hydrology due to new development and social demand for hydrological sciences and sustainable development. Four key scientific issues are identified through systematic analysis and summary of the relative research and international progress, i.e., (1) the long-term dynamic process of socio-hydrological system evolution; (2) quantitative description and driving mechanism analysis of socio-hydrological coupling system; (3) prediction of the trajectories of socio-hydrological system co-evolution, and (4) integrated water resource management from the perspective of water systems. Moreover, opportunities and challenges for developing socio-hydrology are emphasized, including (1) strengthening the research of interdisciplinary theoretical systems; (2) improving and broadening socio-hydrological research technical methods, and (3) supporting integrated water resources management (IWRM) for sustainable utilization goals (SDGs). The review is expected to provide a reference for the future development of socio-hydrology discipline.

Keywords: socio-hydrology; opportunities and challenges; interdiscipline; IWRM; SDGs

1 Introduction

Since the Anthropocene, with industrialization and urbanization, human exploitation of natural resources has put enormous stress on the natural systems. The spatial variability of earth processes has become more intense due to the dual influence of climate change and human activities. The development of the water system has shown an increasingly unpredictable and uncertain trend, resulting in a global water crisis (Pande and Sivapalan, 2017). Traditional hydrology originated from observations and a summary of natural hydrological phenomena (Koutsoyiannis, 2011). The appearance of water-related challenges motivated a

Received: 2021-07-15 **Accepted:** 2022-02-14

Foundation: National Nature Science Foundation of China, No.41890824, No.42101043; Strategic Priority Research Program of the Chinese Academy of Sciences, No.XDA23040304

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paradigm shift in hydrological research, which changed from taking the natural water cycle as the research object (Chow, 1965) to taking into account observations and a summary of natural hydrological phenomena (Koutsoyiannis, 2011).

Based on stable assumptions and regarding human activity as an exogenous of the water cycle, Ehrlich *et al.* (1971) proposed the I=PAT equation to evaluate the impact of human activities, noting that the impact of human activities on the natural environment is related to the population size, consumption level and technological capabilities. Instead of considering one-way effects, Falkenmark (1977) proposed the concept of “hydro-sociology” to explore the two-way feedback between human activities and the water cycle, which was defined as “a science that studies the interaction between humans and water (Falkenmark, 1979). On the basis of previous research results regarding human-water relationship, the integrated water resources management (IWRM) was proposed at the International Conference on Water and the Environment (ICWE) in 1992, which emphasized incorporating the impact of human activities into water resource management (Gorre-Dale, 1992). In terms of research methods, Falkenmark (1997) proposed models to explain and visualize fundamental interactions between human and water and established a conceptual framework including rural and urban water use. In 2004, relying on the Earth System Alliance (EESP), the Global Water System Plan (GWSP) was established (Alcamo *et al.*, 2005). The core task of GWSP is to discover how humans influence the dynamics of the water system and inform decision-makers how to mitigate the environmental and socioeconomic consequences of these effects (Alcamo *et al.*, 2005). Wang *et al.* (2004) coupled natural and artificial water cycles in the “dualistic water cycle” model and developed the dual water cycle theory. Their theory focuses on the co-evolution process of natural and artificial water cycles as well as social driving forces such as population, economics, technology, policy, and government; additionally, water value and water culture were also mentioned. In this period, although many studies had carried out meaningful explorations on human-water interaction, the ignorance of the nonlinear characteristics of human activities made it hard to explain some unintended consequences, such as the irrigation paradox and others (Pande and Sivapalan, 2017). In that case, the paradigm of hydrology research has undergone a further change (Wagener *et al.*, 2010), more disciplines were integrated to explore the dynamic mutual feedback mechanism between society and hydrological systems.

In 2012, Sivapalan *et al.* (2012) introduced the concept of socio-hydrology, which indicates that the study of human-water relationship has entered the third stage, that is, the study of co-evolution. Following this concept, Wei *et al.* (2017) systematically summarized the theories, methods and application of socio-hydrology under the theoretical framework constructed by Sivapalan *et al.* (2012). The International Association of Hydrological Sciences (IAHS), which promotes the development of hydrological sciences to better serve society as its mission (Mao *et al.*, 2017), identified the ten-year science plan that we are currently in as follows: “Panta Rhei-Everything Flows: change in hydrology and society”. The theme aims to better understand the process of controlling the water cycle by emphasizing the dynamics of the water cycle and the rapidly changing human system and improving the ability to predict the dynamics of water resources to support the sustainable development of society in a changing environment (Montanari *et al.*, 2013). The plan is transitioning through the fourth phase (2019–2021), which is named the synthesis phase. The target of this phase is

understanding, estimating and predicting science in practice, and thus revealing the key points of socio-hydrological research. The socio-hydrological development process is shown in Figure 1.

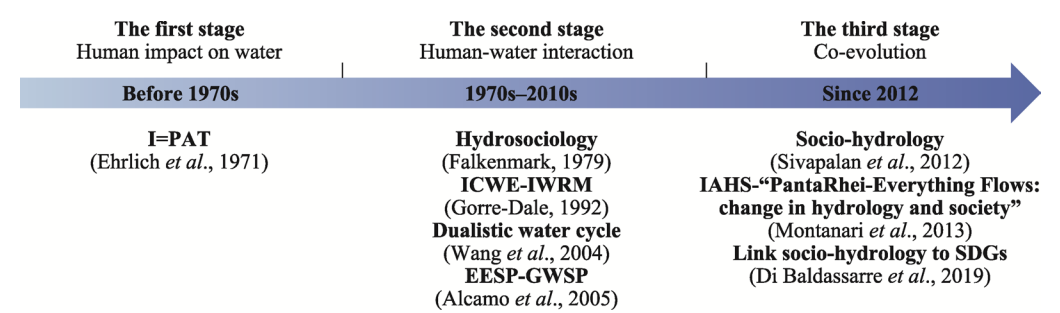


Figure 1 The development process of socio-hydrology

As a discipline supporting water resources management, seeking solutions to water crises has always been the development direction and driving force of socio-hydrology research. Since the beginning of the 21st century, water security issues have become more prominent and humans face severe challenges threatening sustainable development (Braga *et al.*, 2014). The IAHS identified six water crisis hot spots, including ecological degradation, increasing flood risk, water conflict, water quality deterioration, increasing drought severity and groundwater depletion. Di Baldassarre *et al.* (2019) believe that socio-hydrology is able to provide practical scientific support for understanding these phenomena and has great prospects for addressing the sustainable development goals (SDGs). Undoubtedly, the growing water crisis has brought unprecedented opportunities and challenges to socio-hydrology, and in that case, this paper systematically reviews and analyzes the development and current situation of socio-hydrology. By reviewing the development process of socio-hydrology, combing the research status, and discussing key scientific issues, we propose the current opportunities and challenges that socio-hydrology is facing, hopefully providing a meaningful reference for the future development of the discipline.

2 Analysis of socio-hydrology research status

2.1 Bibliometric analysis of literature

To better understand the hot spots in socio-hydrology research, a keyword co-occurrence map of socio-hydrology (Figure 2) is drawn by VOSviewer software (developed by the Center for Science and Technology Studies (CWTS), Leiden University, Netherlands). The data selected is based on the Science Citation Index Expanded (SCI-E) and Social Sciences Citation Index (SSCI) from the Web of Science (WoS) database. In order to obtain the most relevant literature data, the search topic is set as “sociohydrolog* OR socio-hydrolog* OR social hydrolog*” (Wang and Zhang, 2016). The least occurrences of the keyword to be shown in the co-occurrence map are set as 10. According to the results carried out by VOSviewer, there are 2686 keywords in the existing socio-hydrological literature, and 66 keywords meet the threshold. In the co-occurrence map, every single node represents a keyword. The larger the node means the higher the frequency of keyword occurrence, and

the shorter the line segment between two nodes means the closer the correlation between them. As shown in Figure 2, there are five clusters and the core keywords of each cluster are socio-hydrology, management, framework, sustainability, and river-basin respectively. In other words, the above keywords are the frontiers that have received the most attention in socio-hydrology.

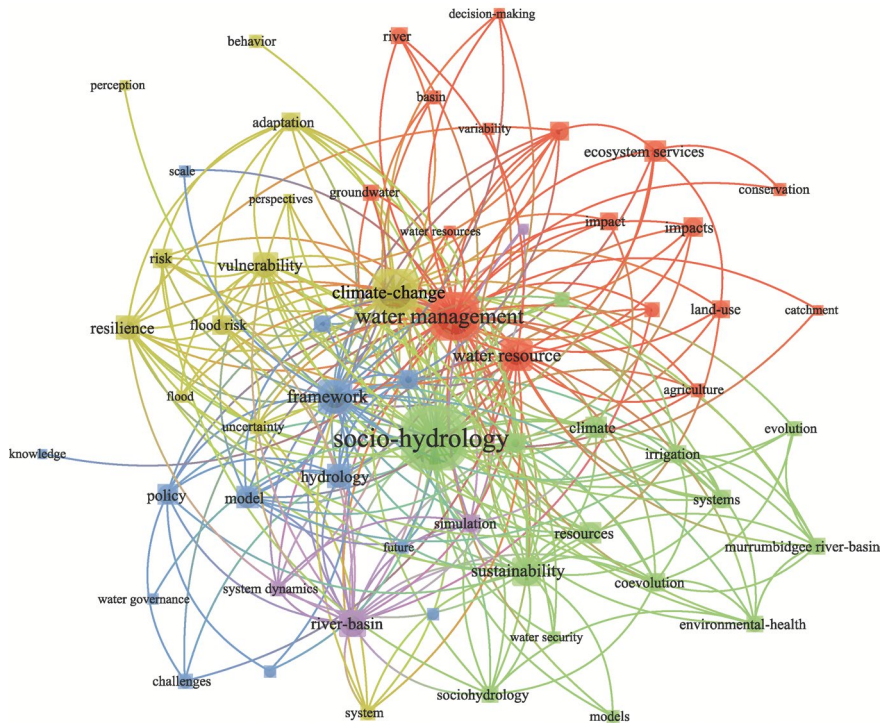


Figure 2 Keyword co-occurrence map of publications in the socio-hydrological field

2.2 Main research content of socio-hydrology discipline

There is a consensus that the emergent water-related issues like irrigation paradox and flooding events have put forward questions for socio-hydrology (Sivapalan *et al.*, 2012; Pande and Sivapalan, 2017). According to Sivapalan *et al.* (2014), socio-hydrology is a use-inspired science and the core aim of socio-hydrology is to give scientific explanations of those phenomena from the perspective of human-water dynamics mutual feedback. Regarding the goal, scientists carry out research mainly from three aspects. The first is the reconstruction and interpretation of the historical evolution of socio-hydrological systems. The time scale of this type of research is normally hundreds or even thousands of years, a lot of them focus on a specific aspect of the human-water relationship, such as water-related values, policies, and technologies and bibliometric methods may be used to analyse the massive text data. The second is the simulation of the human-water mutual feedback process and the exploration of the regular pattern. Socio-hydrological models are often required in this type of research (Sivapalan, 2015). The existing socio-hydrological models can be classified as data-based models, physics-based models and conceptual models according to the way of expression, bottom-up models and top-down models according to the framework structure or

distributed models and lumped models according to whether heterogeneity is considered (Blair and Buytaert, 2016). The models commonly used in socio-hydrology research include agent-based model (ABM), system dynamics (SD), Bayesian networks (BN) etc. The third is management and response to extreme hydrological events and other water crisis issues. Some specific cases may be analysed to explore what role socio-hydrology can play in responding to sustainable development. The representative research examples corresponding to each of the above categories and the methods they have adopted are listed in Table 1.

Table 1 Representative studies of socio-hydrology in the past ten years

Category	Reference	Research object	Approach	Region
Reconstruction and interpretation of the historical evolution of socio-hydrological systems	Lu <i>et al.</i> (2015)	Human-water relationship	Top-down method based on Budyko assumption	Heihe river basin, China
	Xiong <i>et al.</i> (2016)	Water issues	Content analysis	China
	Wei <i>et al.</i> (2015)	Water issues	Content analysis	Australia
	Xiong <i>et al.</i> (2014)	Water issues	Mapping knowledge domain	China
	Zhao <i>et al.</i> (2014)	Water resources management	Historical analysis	China
Simulation of the human-water mutual feedback process and the exploration of the regular pattern	Elshafei <i>et al.</i> (2014)	Socio-hydrology system	SD	Murray-Darling Basin (MDB), Australia
	Aghaie <i>et al.</i> (2020)	Ground water supply	ABM	Rafsanjan, Iran
	Fabre <i>et al.</i> (2015)	Supply-demand dynamics	Component modelling	Herault catchment, France, and Ebro catchment, Spain
	Srinivasan <i>et al.</i> (2015)	Water supply and demand	Multiple-hypothesis model	Chennai, India
	van Dam <i>et al.</i> (2013)	Policy options	BN	Nyando Papyrus wetland, Kenya
	Wu <i>et al.</i> (2019)	Virtual water use	Input-output analysis	Worldwide
	Lopez-Nicolas <i>et al.</i> (2018)	Ground water supply	Water poverty Index	San Luis Potosi, Mexico
	Zhou <i>et al.</i> (2015)	Human-water relationship	Conceptual model	Murrumbidgee catchment, Australia
	Di Baldassarre <i>et al.</i> (2013)	Human-flood interactions	System dynamics	Fictional catchment
	Viglione <i>et al.</i> (2014)	Human-flood interactions	System dynamics	Fictional catchment
Management and response to extreme hydrological events and other water crisis issues	Gunda <i>et al.</i> (2018)	Water stress	Hydrological model and system dynamic model	Valdez acequia, New Mexico
	Kuil <i>et al.</i> (2016)	Drought process	Conceptual model	Ancient Maya
	Sawada and Hanazaki (2020)	Human-flood interactions	Model-data integration	Punjab province of the Indus River basin, Pakistan
	Yu <i>et al.</i> (2017)	Flood resilience	Evolutionary game theory	Coastal Bangladesh
	Van Loon <i>et al.</i> (2019)	Human influence on hydrological droughts	Paired-catchment analysis	Cases in the UK and Australia

2.3 Research gaps in addressing the water crisis

The theory and methodology of socio-hydrology have developed rapidly since it was first proposed. Nevertheless, as an emerging subject, socio-hydrology is still in its infancy and

lacks a complete and mature theoretical system and method system. There are several research gaps existing in addressing the water crisis as follows: (1) there is a lack of a clear mechanism of human-water interaction. Although some studies do make reasonable attempts at exploring the human-water relationship, they are basically based on a holistic macro-framework (Liu *et al.*, 2014) or carry out research from a perspective of traditional hydrology, i.e., they use water balance to describe changes in hydrological and social systems (Zhou *et al.*, 2015), which is difficult to accurately and objectively reflect the systematic interaction between society and hydrology and resulted in the research on the coupling mechanism of socio-hydrological systems is not sufficient. Also, there are other studies considering the process of two-way interactions between social systems and hydrological systems, some are often too simplified and lack an in-depth analysis of the mechanism of co-evolution (Savenije and Van der Zaag, 2008). (2) Much current research is merely qualitative research while the quantitative description is rather vital for model construction and interpretation in socio-hydrological research as it can accurately reveal the co-evolution process of human-water systems and predict the future situation. Therefore, there is a need for more quantitative scientific methods as the primary method of socio-hydrology research (Madani and Shafiee-Jood, 2020). (3) As an interdisciplinary subject involving multiple scales and processes, socio-hydrology often faces the dilemma of lack of data. Socio-hydrological research requires multisource data from hydrological and social systems and appropriate temporal and space scales must be guaranteed, however, researchers always have to face the difficulties in getting enough data and material from different fields (Cotte *et al.*, 2014).

3 Key scientific issues of socio-hydrology inquiry

Sivapalan *et al.* (2014) believed that a theoretical framework of socio-hydrology should be able to help people to explain the past, understand the present, and illuminate sustainable future trajectories of socio-hydrological system co-evolution. Based on that and given the core goals of socio-hydrological science mentioned above, we believe that socio-hydrology inquiry should focus on the following key scientific issues which are oriented to the historical evolution of the socio-hydrological system, the current (or continuous) mutual feedback mechanism, the future development trends and the integrated management of water resources.

3.1 The long-term dynamic process of socio-hydrological system evolution

From a historical perspective, water has played a vital role in developing societies in no matter which culture. Many civilisations' emergence, development, and death are always closely related to water (Sivapalan *et al.*, 2012). Water resource management policies, water conservancy engineering measures, water use methods, irrigation technologies and ways to deal with floods in different historical periods can, to some extent, reflect the level of social development at that period. Therefore, carrying out long-term historical socio-hydrological research on the time scale of hundreds or even thousands of years is undoubtedly necessary.

Socio-hydrology evolution includes changes in both the social and hydrological system and the dynamic process of their mutual feedback. Those processes can take a relatively long

time due to a time lag in the interaction between hydrological and social systems. For instance, changes in land use (such as urbanization, agricultural expansion, or intensification) always have long-term effects on erosion and sediment transfer rates and nutrient fluxes (Kalnay and Cai, 2003). Additionally, large-scale water diversion (such as water storage, hydroelectric power generation, and irrigation projects) may affect the hydrological process and water quality of inland waters for decades to hundreds of years (Qiu and Zhu, 2013). What's more, agricultural, aquaculture, and industrial pollution sources that directly or indirectly pollute water bodies (such as the release of chemical substances, fertilizer use, sewage and sludge treatment, and atmospheric deposition) may have legacy effects on water bodies for decades or even longer (Huang *et al.*, 2004). Besides, the natural system itself is changing slowly with a hysteresis effect (Parmesan and Yohe, 2003). So a long timescale is quite necessary for historical socio-hydrological analysis to uncover the evolution trajectory of each sector and agent in the whole system.

Faced with the fact that historical data are often scarce, hydrologic reconstruction has become a key to historical socio-hydrological research, as it can provide long-term data on hydrological variables and other related environmental variables, including human activities (Montanari *et al.*, 2013). The proxy data material for hydrological reconstruction includes tree rings, lake deposits, ice cores, pollen, historical documents, etc. By using proxy data, climate change data such as runoff, precipitation, temperature and extreme hydrological events (Mischke *et al.*, 2003; Sivakumar *et al.*, 2007; Razumovskii and Gololobova, 2008; Sivakumar, 2009; Molotch *et al.*, 2015) as well as human activity data such as residential areas, the establishment of water conservancy projects and farmland reclamation (Karmanov *et al.*, 2011) can be produced. With the development of geographic information system (GIS) and remote sensing (RS) technology, studies using a combination of historical literature and modern technology have been widely carried out in the field of human geography, enabling the use of historical population surveys, demographic statistics, and population migration memory population spatial distribution analysis to be realized (McLeman *et al.*, 2010).

In summary, carrying out socio-hydrological research on a scale of hundreds or even thousands of years is of great significance to understanding the dynamic process of the co-evolution of socio-hydrological system. Hydrologic reconstruction provides a basis for understanding the evolutionary laws of the natural environment on different scales and its relationship with human activities; thus, it serves as the premise for revealing the evolutionary mechanism of a long-term human-water coupling system. It is foreseeable that the future research direction will be not only the description of the evolution process of human-water system but also the detailed analysis of the mutual feedback process so as to gain experience and enlightenment from history to improve the current water resource management and future prediction level.

3.2 Quantitative description and driving mechanism analysis of socio-hydrological coupling system

Quantitative research is as important as qualitative to socio-hydrology as it enables researchers to test hypotheses, simulate systems, and predict the future development trajectory of the system. The content analysis method is a common quantitative analysis method that reveals trends and structural patterns from text data (Stockwell *et al.*, 2009) which can qual-

itatively and quantitatively analyse text information (Insch *et al.*, 1997) to provide an effective tool for tracking changes in media reports (Kirilenko *et al.*, 2012). Wei *et al.* (2015) used the content analysis method to quantitatively analyse the evolution of water issues in Australia through newspaper coverage. Whaley and Weatherhead (2016) conducted a directed content analysis of seven key water policy documents of England to investigate the contribution English water policy can make for adaptive co-management. Another quantitative method which is becoming increasingly popular is mapping knowledge domain. It is an emerging cross-cutting frontier that spans the fields of science, information science, scientometrics, computer science, and applied mathematics. Mapping knowledge domain includes a process of mining, analysing, classifying, navigating and visualizing (mapping) knowledge (Shiffrin and Börner, 2004) that can reveal the structural relationship and evolution process of a knowledge field. The approach of mapping knowledge domain is applied to similar research objects and employs similar information carriers with content analysis but is more dynamic, intuitive, and adaptable to the era of information explosion. As an emerging big data technology, the application of mapping knowledge domain in socio-hydrology implies the great potential of big data in socio-hydrological research.

The development of quantitative description methods has made the uncovering of the driving mechanism of the socio-hydrological coupling systems more likely to come true. Also, there are many existing studies on coupling systems of nature and society inspiring the research on the driving mechanism of socio-hydrological coupling system. Coupled human and natural systems (CHANS) (Liu *et al.*, 2007) and Socio-ecological systems (SEs) (Rogachev *et al.*, 2017) are two topics closely related to socio-hydrology as they both take the keywords in socio-hydrology research like resilience (Uribe-Castaneda *et al.*, 2018), policy (Konar *et al.*, 2016) or co-evolution (Ferraro *et al.*, 2019) into thinking around and thus have similar research issues (Blair and Buytaert, 2016). It can be speculated that there is a general thinking that applies to all of the above studies. Socio-hydrology should gain experience from the theoretical systems and technical methods of other studies or disciplines (Seidl and Barthel, 2017; Wesselink *et al.*, 2017) and strengthen the quantitative study of the two-way feedback process between society and hydrological system to better understand and interpret the human-water coupling system.

3.3 Prediction of the trajectories of socio-hydrological system co-evolution

A key role of socio-hydrology is to support the formulation of water resources management policies (Melsen *et al.*, 2018). To achieve the goal, scientific and accurate quantitative prediction of the co-evolution trajectories of the socio-hydrological system is of vital importance. The goal of prediction is to let decision-makers understand the consequences of the proposed decision, whether it is irreversible, and how to incorporate new information to make the decision more adaptive. Conventional prediction is always based on static scenario assumptions that assume the factors in the social system are in a stable state during the study period (Fiseha *et al.*, 2014). However, the nonlinear feedback characteristics of human activities are always overlooked which has created obstacles to the prediction of human-water systems co-evolution. It is well-known that factors such as people's values, beliefs, education levels, etc. have significant effects on people's behavioural characteristics, resulting in changes in policies and systems, which might, in turn, affect individual behaviours

(Srinivasan *et al.*, 2017). The above processes are complex and full of uncertainty (Yu and Rockstrom, 2015) and may result in emergent behaviour and unintended consequences (McMillan *et al.*, 2016). Faced with the difficulties in socio-hydrological prediction, Srinivasan *et al.* (2017) argued that stakeholder participation should be promoted and the local scale should be integrated with the global scale when appropriate. In addition, socio-hydrological models should be able to interpret adaptive responses by different sectors to simulate unexpected and emergent situations, although this is clearly challenging. There are other excellent papers proposing some solutions for how to better model socio-hydrological prediction in detail (Troy *et al.*, 2015; Blair and Buytaert, 2016), and we highly recommend them.

In addition to modelling, we believe that big data has a great potential in socio-hydrological prediction. Big data technology enables the transformation of a very difficult prediction problem into a relatively simple description problem, which is beyond the reach of traditional small data sets. Also, big data methods are in line with characteristics of the Internet era and more easily describe and capture collective human activities in geographic space through social media networks (Ren *et al.*, 2019). For example, Moat *et al.* (2014) discussed how to use big data to predict collective behaviour in the real world, Jiang and Ren (2019) predicted the living structure of human activities by using big data. In fact, the development of big data prediction is relatively mature in the field of environmental change prediction (Soltis and Soltis 2016). Chang *et al.* (2018) established a big data access framework to realize the analysis of air quality. Athey (2017) discussed how to use big data to address policy problems on the basis of machine learning. Based on the above research, we can reasonably believe that there are bright prospects for big data in socio-hydrological prediction.

3.4 Integrated water resource management from the perspective of water systems

The water-related challenges involving climate, ecology, environment, and society in a changing world are complex, often manifested as water shortages, water pollution, water ecological crises, water disasters, and water management problems. As mentioned before, GWSP contributes to clarifying the control factors of the water cycle as well as surface processes at different temporal and spatial scales based on the multi-circle evolution and revealing the impact of human activities and climate change on land surface water volume, water quality, environment, and social economy. What is more, GWSP provides a scientific basis for reducing flood and drought disaster risk, supporting the sustainable use of water resources, ecological protection and water pollution control, etc. Water governance under the guidance of GWSP should include not only the regulation of water resource engineering construction but also the regulation of the legal system of water and institutional construction, water economy and water price regulation as well as water management system innovation. Undoubtedly, the water system provides new ideas for addressing water-related challenges. According to the GWSP, Xia *et al.* (2018) believed that the core process of water problems is the water cycle, and at the same time biological and biogeochemical processes as well as humanistic process of social and economic development and management are also involved. Water resources management from the perspective of water system is the key to solving the problems. Referring to the structure of the natural water cycle proposed by GWSP, a social water cycle is shown in Figure 3 to illustrate the sectors and processes in social water cycle.

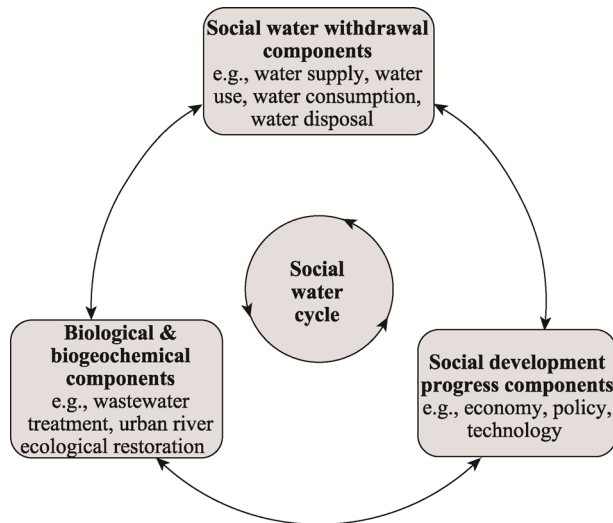


Figure 3 Social water cycle from the perspective of the water system

4 Opportunities and challenges for socio-hydrology development

4.1 Strengthening the research of interdisciplinary theoretical systems

Socio-hydrology intersects with many other disciplines or theories. For example, complex system science is often applied to socio-hydrological modelling (Liu *et al.*, 2007), and hydro-economics focuses on the issue of water resource allocation, which is seen as a key determinant of the interactions between social and natural systems (Lu *et al.*, 2018). Sociology systematically studies human social behaviour (Portes, 1998), including population and social groups, social systems, behavioural norms, and society evolution (Bottomore, 2010). Those disciplines provide theoretical support for the social side of research on socio-hydrological systems.

McCurley and Jawitz (2017) believed that combining other disciplines that were formerly exotic for hydrology might promote the generation of new insights and provide different perspectives to answer old questions. As mentioned previously, developing interdisciplinary theory is a vital method for understanding the interaction relationship between humans and water systems in this rapidly changing world (Seidl and Barthel, 2017; Wesselink *et al.*, 2017) since it can lead to a better understanding of human-water interactions, help to correct assumptions and address uncertainties (Di Baldassarre *et al.*, 2016). In the past, the research and modelling of socio-hydrology were mainly carried out by hydrologists, which might cause the problem of oversimplifying the description and analysis of the social sectors. In recent years, the problem is gradually becoming more aware and the research trend has moved to exploring how to combine natural sciences and social sciences with the aim of applying interdisciplinary methods to socio-hydrology, including a comparison of the principles of action between different disciplines and a delineation of the cooperation scope. Wesselink *et al.* (2017) argued that plentiful social research cases related to water could provide a basis for conceptual and quantitative modelling of socio-hydrology. Massuel *et al.* (2018) proposed an interdisciplinary socio-hydrological research method that expanded of

analysis and modelling of the water system and complemented traditional hydrological methods, the method provides people with a deeper understanding of the mechanism behind problems related to water and inspires debates on the interactions between social and political decisions.

It is foreseeable that with the development of the understanding on the relationship between human and water, more other disciplines will be creatively combined with socio-hydrology. The exploration and development of interdisciplinary methods will undoubtedly be opportunities and challenges for socio-hydrology discipline.

4.2 Improving and broadening socio-hydrological research technical methods

How to solve the difficulties in socio-hydrological modelling and improve the scientificity and accuracy of the model is an urgent challenge faced by socio-hydrology (Blair and Buytaert, 2016). Regarding the bidirectional interactions between society and water, there have been a lot of attempts. For example, based on the water system theory mentioned in 3.4 and research related to urban water system simulation, Zhang *et al.* (2021) proposed a theoretical framework for the urban water system. In the framework, the socio-economic water withdrawal process module of the model considers the impact of policy, technology, and society on the water consumption and drainage of various industries. This consideration quantifies the influence of technological factors and policy factors on the evolution of the model and reveals the coupling driving mechanism of urban water-society-environment.

Generally speaking, the research on the socio-hydrological mutual feedback mechanism often faces difficulties from several aspects. First, human variables are quite complex and full of uncertainty when the prediction model involves social, economic and political systems (Wagener *et al.*, 2010). Second, considering the long time and large spatial scale of socio-hydrology, it is difficult to accumulate enough data to run, calibrate, and validate a model (Blair and Buytaert, 2016), the observations may not be sufficient to support predictions of changing environments. Third, there is always a time lag of the social impact on the natural system, which makes the essence of the related water problems can be easily concealed by complex dynamic factors. Fourth, human society is changing rapidly, while the changes in the natural environment are rather slow; when different data are input together into an integrated model, the coordination of the spatial and temporal scales can be difficult (Blomquist and Schlager, 2005). In addition, a variety of complexities such as algorithm complexity, deterministic complexity and aggregate complexity also exist (Manson, 2001) when applying the modelling method. In that context, the multisource data utilization capabilities of models need to be improved, and the detailed description of hydrological processes that are driven by social forces need to be strengthened to make simulations of the social and hydrological processes coupling more accurate. It is noted that the explosive development of big data has enabled it to be applied in social science, policy, environment, water conservancy and many other fields (Chen *et al.*, 2014). Based on the characteristics of big data technology and referring to its application experience in other fields, we believe that big data can also contribute to broadening the ways to obtain socio-hydrological data (Pei *et al.*, 2020), finding laws from complex socio-hydrological processes (Wang and Zhang, 2016) and visualizing the results (van Eck and Waltman, 2009).

4.3 Supporting integrated water resources management (IWRM) for sustainable utilization goals (SDGs)

Indeed, from a global perspective, the current water crisis facing humans is essentially a water resource management crisis (Voeroesmarty *et al.*, 2010). There is an urgent need for scientific and effective water resource management plans since strengthening the sustainable management of catchment water resources is considered a fundamental measure to address the water crisis. IWRM is supposed to provide solutions to address water crisis and reach SDGs by coordinating the development of water resources and the social economy while also considering the effect on the ecological environment. Xia *et al.* (2017) constructed a framework of integrated urban water systems, reflecting the interrelationships among hydrological components, water pollution components, and water management components. In addition, the IWRM applied in the Murray Darling Basin in Australia has provided an advanced experience in catchment management (Xia *et al.*, 2009). There is a consensus that water resource management must shift from natural hydrological concepts to socio-hydrological concepts, which requires the support of socio-hydrological theory and methods. The process of socio-hydrology supporting the IWRM for SDGs is described in Figure 4. Briefly, the emergence of water crises poses questions to socio-hydrology, which requires socio-hydrology to initially understand the reasons for these phenomena. And then, to alleviate or solve water crises, IWRM provides solutions based on the causes of the water crises. Socio-hydrology is expected to analyse the influence mechanism of management measures and provide theoretical and technical support for IWRM as well as predict the possible impact and results of IWRM. After the measures are implemented, socio-hydrology also needs to evaluate the actual effects to guide the next adjustment and improvement. In addition, aiming to achieve the targets of SDGs, socio-hydrology should broaden its scope from floods, droughts and other conventional water-related issues to food, energy production and human health, which are included in SDGs (Herrera, 2019).

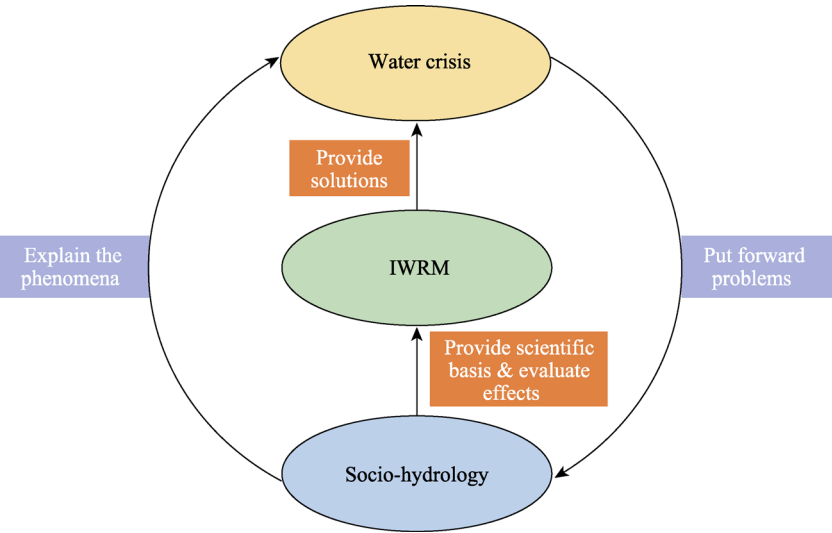


Figure 4 The process of socio-hydrology supporting IWRM for SDGs

5 Concluding remark

In this paper, we started from the shifts in the hydrology research paradigm and conducted a literature review of socio-hydrology. We argued that the discipline of socio-hydrology has experienced a slow to rapid development process but has not yet reached a mature stage. The understanding of the relationship between human and water has changed from seeing the human impact as an exogenous factor to the exploration of the internal dynamic mutual feedback mechanism. In view of the connotation of socio-hydrology under the current research background, further research on four key scientific issues is needed: the long-term dynamic process of socio-hydrological system evolution, quantitative description and driving mechanism analysis of socio-hydrological coupling system, prediction of the trajectories of socio-hydrological system co-evolution and IWRM from the perspective of water systems. These key scientific issues explore the dynamic feedback relationship and coupling mechanism between social and hydrological systems from the three perspectives of history, the present and the future and help people more comprehensively and deeply understand the human-water coupling system. And based on the key scientific issues, we propose the opportunities and challenges that socio-hydrology is facing: (1) Strengthening interdisciplinary research theoretically and making the disciplinary system of socio-hydrology more perfect. (2) Improving and broadening socio-hydrological research technical methods. Appropriate alternative methods, including big data, should be considered to combine with the model or to become an option. (3) Enhancing the practicability of the discipline, increasing the support for IWRM to meet water crisis problems, and exploring the future of the discipline in practice.

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